ModelIt!

Student Activity Pack

What Defines the Ethics of Science?

Modeling Scientific Responsibility, Dual-Use Research, and Evidence-Based Policy

Grade 12L3 | NGSS HS-ETS1-3 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Can you think of a scientific discovery that has been used for both tremendous good and terrible harm? Describe the discovery and both applications.**

2. **Who do you think should decide which scientific research gets funded and which gets restricted? Scientists? Politicians? The public? Explain your reasoning.**

3. **What does scientific integrity mean to you, and what happens to society when people stop trusting science?**

4. **Is there any scientific question that should NOT be pursued, even if answering it would advance human knowledge? Why or why not?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Research Funding Allocation | Dual-Use Potential | Public Trust in Science | Institutional Oversight Rigor | Publication Transparency | Researcher Ethical Awareness | Benefit Distribution Equity | Risk Distribution Equity | Innovation Rate | Societal Impact Score

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? **Explain ONE relationship in your model. What happens when one component changes?**

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Standard Research Environment

Dual-Use: Low | Funding: Moderate | Oversight: Standard | Trust: High

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Dual-Use Crisis

Dual-Use: Extreme | Oversight: Maximum | Publication: Restricted | Public Trust: Variable

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Trust Collapse

Public Trust: Very Low | All other components unchanged

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

International Governance Coordination — The degree to which ethical standards and research restrictions are harmonized across nations — if one country restricts dangerous research but another does not, the research simply moves to the less regulated jurisdiction, making international coordination essential for effective governance of dual-use research

Intellectual Property Framework — The system of patents, copyrights, and trade secrets that determines who owns scientific knowledge and who can access it — proprietary restrictions can accelerate commercialization but also limit access, creating tension between innovation incentives and equitable distribution of scientific benefits

Whistleblower Protection — The legal and institutional protections available to researchers who report ethical violations, data fabrication, or safety concerns — effective whistleblower protection is essential for scientific self-correction but is often inadequate, leaving reporters of misconduct vulnerable to retaliation

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Science has given humanity vaccines, electricity, computers, and the ability to feed 8 billion people. It has also given us nuclear weapons, biological warfare agents, surveillance technology, and the tools for ecological destruction. Who decides which research should be pursued, who benefits from scientific progress, and what happens when the same knowledge can both save and destroy lives? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that maximum Innovation Rate does not maximize Societal Impact Score, and what does this imply about the relationship between scientific freedom and scientific responsibility?**

2. **What does the public trust scenario reveal about the feedback loop between transparency, trust, funding, and innovation — and why is trust so difficult to rebuild once lost?**

3. **How does the dual-use scenario illustrate the fundamental tension between sharing knowledge openly (which enables progress and scrutiny) and restricting knowledge (which prevents misuse but also prevents beneficial applications)?**

4. **How would adding international governance coordination to your model change the predicted effectiveness of national-level research restrictions?**

5. **What are the limitations of modeling scientific ethics with quantitative components when ethical decisions ultimately involve value judgments that cannot be reduced to numbers?**

8 STEM Challenge: Design a Science Ethics Governance Framework

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YOUR ENGINEERING MISSION

Design a comprehensive governance framework for a national research institution that maximizes beneficial innovation while preventing dual-use misuse, ensuring equitable distribution of benefits and risks, and maintaining public trust in science.

THE SCENARIO

A new national research institute is being created to pursue frontier science in artificial intelligence, synthetic biology, and climate engineering — three fields with enormous beneficial potential and serious dual-use concerns. The institute will receive $5 billion annually in public funding. Your ethics advisory board must design the governance framework: ethical review processes, publication policies, dual-use research oversight, public engagement mechanisms, equity mandates, and accountability structures. The framework must enable rapid innovation while preventing research from causing disproportionate harm to any community.

Engineering Design Thinking

1. WHAT IS THE STRUCTURE AND COMPOSITION OF YOUR ETHICS REVIEW BOARD, AND HOW DOES IT MAINTAIN INDEPENDENCE FROM RESEARCHERS AND FUNDERS?

2. WHAT SPECIFIC CRITERIA DOES YOUR FRAMEWORK USE TO CLASSIFY RESEARCH AS DUAL-USE AND WHAT RESTRICTIONS APPLY AT EACH CLASSIFICATION LEVEL?

3. HOW DOES YOUR FRAMEWORK ENSURE THAT COMMUNITIES AFFECTED BY RESEARCH — BOTH POSITIVELY AND NEGATIVELY — HAVE A VOICE IN RESEARCH GOVERNANCE?

4. WHAT ACCOUNTABILITY MECHANISMS DOES YOUR FRAMEWORK INCLUDE WHEN RESEARCH CAUSES UNINTENDED HARM DESPITE ETHICAL REVIEW?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Research ethicists and bioethicists who evaluate the moral implications of scientific research earn $65,000-$130,000/year at universities, hospitals, and government agencies. Science policy advisors who translate research findings into government policy earn $70,000-$140,000/year. IRB administrators who manage institutional ethics review processes earn $55,000-$95,000/year. Science communication specialists who build public trust through transparent engagement earn $50,000-$100,000/year.